Applications of BST

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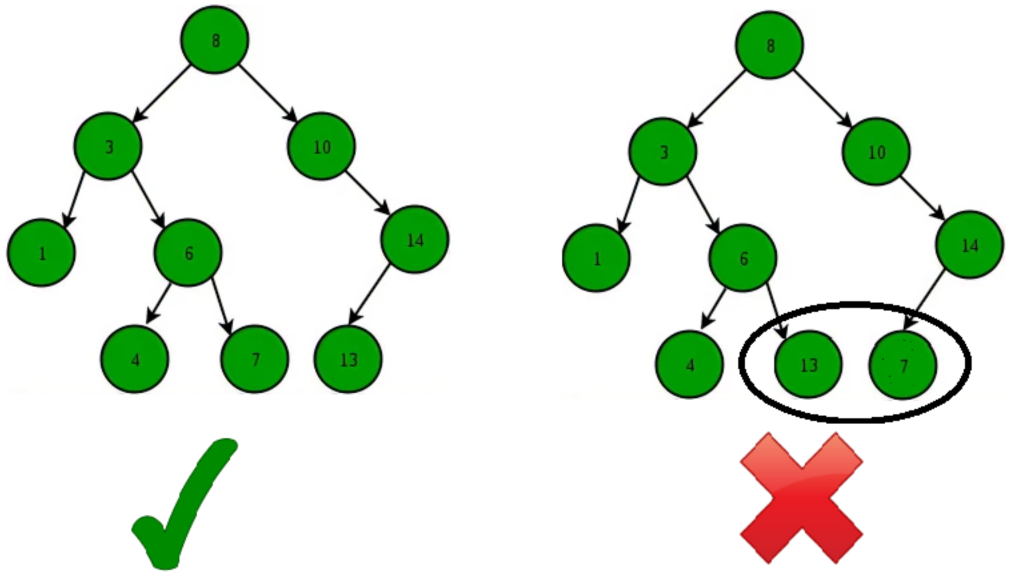
Binary Search Tree (BST) is a data structure that is commonly used to implement efficient searching, insertion, and deletion operations along with maintaining sorted sequence of data. Please remember the following properties of BSTs before moving forward.

- The left subtree of a node contains only nodes with keys lesser than the node's key.
- The right subtree of a node contains only nodes with keys greater than the node's key.
- The left and right subtree each must also be a binary search tree. There must be no duplicate nodes.



Binary Search Tree - Structure

A BST supports operations like search, insert, delete, maximum, minimum, floor, ceil, greater, smaller, etc in O(h) time where h is height of the BST. To keep height less, self balancing BSTs (like AVL and Red Black Trees) are used in practice. These Self-Balancing BSTs maintain the height as O(Log n). Therefore all of the above mentioned operations become O(Log n). Together with these, BST also allows sorted order traversal of data in O(n) time.

1. A Self-Balancing Binary Search Tree is used to maintain sorted stream of data. For example, suppose we are getting online orders placed and we want to maintain the live data (in RAM) in sorted order of prices. For example, we wish to know number of items purchased at cost below a given cost at any moment. Or we wish to know number of items purchased at higher cost than given cost.
2. A Self-Balancing Binary Search Tree is used to implement doubly ended priority queue. With a Binary Heap, we can either implement a priority queue with support of extractMin() or with extractMax(). If we wish to support both the operations, we use a Self-Balancing Binary Search Tree to do both in O(Log n)
3. There are many more algorithm problems where a Self-Balancing BST is the best suited data structure, like count smaller elements on right, Smallest Greater Element on Right Side, etc.
4. A BST can be used to sort a large dataset. By inserting the elements of the dataset into a BST and then performing an in-order traversal, the elements will be returned in sorted order. When compared to normal sorting algorithms, the advantage here is, we can later insert / delete items in O(Log n) time.
5. Variations of BST like B Tree and B+ Tree are used in Database indexing.
6. **TreeMap** and **TreeSet in** Java, and set and map in C++ are internally implemented using self-balancing BSTs, more formally a Red-Black Tree.

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Given a BST, the task is to insert a new node in this BST.Example: How to Insert a value in a Binary Search Tree:A new key is always inserted at the leaf by maintaining the property of the binary search...

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Given a BST, the task is to search a node in this BST. For searching a value in BST, consider it as a sorted array. Now we can easily perform search operation in BST using Binary Search Algorithm. Input: Root of...

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Deletion in Binary Search Tree (BST)

Given a BST, the task is to delete a node in this BST, which can be broken down into 3 scenarios:Case 1. Delete a Leaf Node in BSTDeletion in BSTCase 2. Delete a Node with Single Child in BSTDeleting a...

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Balance a Binary Search Tree

Given a BST (Binary Search Tree) that may be unbalanced, the task is to convert it into a balanced BST that has the minimum possible height.Examples: Input: Output: Explanation: The above unbalanced BST...

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